#Histogram

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
plt.figure()
plt.hist(df['Rating'], bins=20)
plt.title("Histogram of Movie Ratings")
plt.xlabel("Rating")
plt.ylabel("Frequency")
plt.show()
```

#Line chart

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
plt.figure()
yearly_avg = df.groupby('Year')['Rating'].mean()
plt.plot(yearly_avg.index, yearly_avg.values)
plt.title("Average Rating Over Years")
plt.xlabel("Year")
plt.ylabel("Average Rating")
plt.show()
```

#Pie chart

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
plt.figure()
```

```
df['Era'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title("Movies Distribution by Era")
plt.ylabel("")
plt.show()
```

#Box Plot

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
plt.figure()
sns.boxplot(x=df['Runtime(Mins)'])
plt.title("Runtime Distribution")
plt.show()
```

#Scatter plot

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
plt.figure()
plt.scatter(df['Runtime(Mins)'], df['Rating'])
plt.title("Runtime vs Rating")
plt.xlabel("Runtime")
plt.ylabel("Rating")
plt.show()
```

#Heatmap

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
plt.figure()
corr = df[['Year', 'Rating', 'Runtime(Mins)', 'Movie_Age']].corr()
sns.heatmap(corr, annot=True)
plt.title("Correlation Matrix")
plt.show()
```

#Pairplot

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
sns.pairplot(df[['Year', 'Rating', 'Runtime(Mins)', 'Movie_Age']])
plt.show()
```

#Bubble plot

```
#import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
plt.figure()
plt.scatter(
    df['Runtime(Mins)'],
    df['Rating'],
    s=df['Rating'] * 20,
    alpha=0.5
)
```

```
plt.title("Bubble Plot: Runtime vs Rating")
plt.xlabel("Runtime")
plt.ylabel("Rating")
plt.show()
```

#KDE plot

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
plt.figure()
sns.kdeplot(df['Rating'], fill=True)
plt.title("Rating Distribution (KDE)")
plt.show()
```

#Hypothesis testing

```
import pandas as pd
from scipy.stats import pearsonr
# Load dataset
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
# Remove missing values
df = df.dropna()
# Perform Pearson correlation test
corr, p_value = pearsonr(df['Runtime(Mins)'], df['Rating'])
print("Correlation:", corr)
print("P-value:", p_value)

# Decision
if p_value < 0.05:
```

```
    print("Reject H0 → Runtime significantly affects Rating")
else:
    print("Fail to Reject H0 → No significant relationship")
```

#Statistical Modeling

```
import pandas as pd
from sklearn.model_selection import train_test_split
# Load dataset
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
# Remove missing values
df = df.dropna()
# Define variables
X = df[['Runtime(Mins)']]   # Independent variable
y = df['Rating']            # Dependent variable
# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
print("Training Data Size:", X_train.shape)
print("Testing Data Size:", X_test.shape)
```

#Linear Regression Model

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_absolute_error
from sklearn.model_selection import train_test_split

# Load dataset
```

```
df = pd.read_csv("final_IMDb_dataset_with_lead_actor.csv")
# Create model
model = LinearRegression()
# Train model
model.fit(X_train, y_train)
# Predict
y_pred = model.predict(X_test)

# Evaluation
print("R2 Score:", r2_score(y_test, y_pred))
print("MAE:", mean_absolute_error(y_test, y_pred))
print("Intercept:", model.intercept_)
print("Slope:", model.coef_)

# Visualization
plt.scatter(df['Runtime(Mins)'], df['Rating'])
plt.plot(df['Runtime(Mins)'], model.predict(X), color='red')
plt.xlabel("Runtime")
plt.ylabel("Rating")
plt.title("Linear Regression Model")
plt.show()
```